

VIJAY YADAV

PhD | Numerical Simulations | Scientific Machine Learning (SciML)

🇮🇳 Indian [ivijayyadav.github.io](https://github.com/ivijayyadav) [@yadav.59@iitj.ac.in](mailto:yadav.59@iitj.ac.in) 📄 Publications [0009-0006-4903-8044](https://orcid.org/0009-0006-4903-8044) 📍 Jodhpur, IN

EDUCATION

PhD, Physics

Indian Institute of Technology Jodhpur

📅 28/12/2021 – Ongoing

📍 Rajasthan, India

- PhD advisor: Prabhat K. Jaiswal
- Thesis title: Accelerating numerical simulations using machine learning.

Master of Science, Physics

National Institute of Technology Tiruchirappalli

📅 01/08/2018 – 30/06/2020

📍 Tamil Nadu, India

- Thesis title: CuS- rGO nanocomposites for gas sensing applications, Grade: 9.0
- Thesis advisor: S. Manivannan
- Final grade: 7.8

Bachelor of Science, Physics

University of Delhi

📅 20/07/2015 – 09/07/2018

📍 Delhi, India

- Dissertation title: Review on efficiency enhancement techniques for solar cell, Grade: 10.0
- Thesis advisor: Mukesh Kumar
- Final grade: 8.0

PUBLICATIONS

Published

1. [V. Yadav](#), P. K. Jaiswal, R. Podgornik, and S. Kumari, "Charge Regulation Effect on Nanoparticle Interaction Mediated by a Polyelectrolyte," *Macromolecules* **59**, 11, 6398-6409 (2026). [\[Selected for Front Cover\]](#)
2. [V. Yadav](#), M. Priya, M. D. Shrimali, and P. K. Jaiswal, "Graph neural network for prediction of phase-ordering kinetics," *Chaos* **35**, 061101 (2025).
3. S. Chauhan, S. Mandal, [V. Yadav](#), P. K. Jaiswal, M. Priya, and M. D. Shrimali, "Machine learning based prediction of phase ordering dynamics," *Chaos* **33**, 061103 (2023).

Submitted

1. [V. Yadav](#), M. Priya, M. D. Shrimali, and P. K. Jaiswal, "Physics-guided convolutional neural network for domain growth prediction in systems with conserved kinetics," (2026).
2. [V. Yadav](#), S. Kumari, and P. K. Jaiswal, "Machine learning accelerated simulation of charge regulation," (2026).

In Preparation

1. [V. Yadav](#), M. Priya, M. D. Shrimali, and P. K. Jaiswal, "Machine learning accelerated simulation of surface directed spinodal decomposition," (2026).

CONFERENCE PRESENTATIONS

- 05/2026: Poster presentation at the 10th International Soft Matter Conference on "Data-Driven Modelling of Phase-Ordering Kinetics."
- 03/2026: Oral talk at the APS Global Physics Summit on "Data-Dr"

AWARDS

- 2026: Publication “Charge Regulation Effect on Nanoparticle Interaction Mediated by a Polyelectrolyte” selected as the front cover of *Macromolecules*.
- 2023 – 2026: Senior Research Fellowship (SRF) from the Ministry of Education, India.
- 2022 – 2023: Junior Research Fellowship (JRF) from the Ministry of Education, India.
- 2017: Won first prize for the project “Drone Weather Station” during the Golden Jubilee function of SSN College, University of Delhi.

SKILLS

Programming Languages

Python

Mojo

Simulation Tools/Techniques

LAMMPS

Finite Difference Method

Machine Learning Frameworks

PyTorch

Scikit-Learn

Visualization Tools

OVITO

Matplotlib

Additional Skills

SLURM Cluster

Git

Jupyter

Marimo

LANGUAGES

English

Fluent

Hindi

Mother tongue

REFEREES

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